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HD 170312 is a bright, long-period eclipsing binary

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ABSTRACT

I identify HD 170312 ($V = 8.7$), a star with an evolved F primary, as a long-period eclipsing binary. Its TESS light curves show four eclipses centered on the star, two 8.5 percent deep and two 15.4 percent deep; the fourth lies in cadences flagged for scattered light. Only two orbital periods fit them, 137.90 and 68.95 d. A possible egress at the start of sector 101, an eclipse in a NEOWISE visit of 2014 and ASAS-3 photometry favor 68.95 d. A deeper eclipse at BTJD 4475.8152 (2027 March 11), in TESS sector 114, would settle the period.

Keywords: Eclipsing binary stars (444) — Stellar photometry (1620) — Light curves (918)

I report an eclipsing binary that had not been recognized: HD 170312 (TIC 292292315; F3 IV/V, $V = 8.71$; Stassun et al. 2019). No catalog I searched lists it as variable, including the TESS eclipsing binary catalog (Prša et al. 2022) and the Gaia DR3 variability tables (Gaia Collaboration 2016, 2023). I call the F star A and its companion B.

The star was found by a search of multi-sector TESS 2-minute light curves (Ricker et al. 2015; Jenkins et al. 2016) with transitgpu 0.4.0 (Meneguzzo 2026). The search peak, 12.9 d, is not the orbital period. It is 1792.694/139, a spacing that places the three eclipses of unflagged data on one ephemeris. SPOC's searches triggered on these eclipses seven times, and none became a TESS Object of Interest.

TESS recorded four complete eclipses (Figure 1b). Three lie in unflagged data, in sectors 39, 102 and 103, at BTJD 2377.1543 ± 0.0002, 4131.0663 ± 0.0001 and 4169.8482 ± 0.0005. They are of two kinds: the first and third are 8.5 percent deep, the second 15.4 percent, each uncertain by about 1.5 percentage points because the PDC correction tilts the baseline beside the sector 102 eclipse.

The fourth eclipse lies in data that the pipeline sets aside. Sector 93 continues for a day after its last unflagged cadence, in 715 flagged cadences, 359 of them for scattered light, which PDCSAP omits. Their SAP flux holds a deep eclipse. Fitted with the shape of the sector 102 eclipse, it is centered at BTJD 3855.2675, 20 s from the predicted time, at 1.003 times the depth of that eclipse. Two tests show that it is the star's and not the scattered light's. The fall in each of 121 pixels is proportional to that pixel's light out of eclipse (correlation −0.998). The depth changes by under 0.1 percentage point across five estimates of the background, which rises more than tenfold over the eclipse.

The eclipses come from the HD 170312 system. The dimming lies within 2.2″ of A, and every Gaia DR3 source within 5″ is $G = 19.6$ or fainter, 11 magnitudes too faint to make a 15 percent dip.

The eclipses also show that B is a star, too large for a planet or a brown dwarf. I do not know which eclipse is B passing in front of A, so the shallow eclipse bounds the size of B: $R_B \geq R_A\sqrt{\delta/c}$, where $c = 1.134$ is the ratio of the central to the mean intensity of A for quadratic limb darkening at 6645 K and $\log g = 3.5$ (Claret 2017). This gives $R_B \geq 0.95\,R_\odot$ with the TIC radius $R_A = 3.46\,R_\odot$. Models of two eclipsing stars fitted to the eclipses need A to be 24 to 34 percent smaller than its TIC radius, which I cannot explain. With their smallest radius of A, $2.27\,R_\odot$, the bound is $R_B \geq 0.62\,R_\odot$.

Two periods fit the four eclipses. The shallow eclipses are $\Delta T = 1792.694$ d apart, so each kind recurs with $P = \Delta T/N$ for an integer N. Seven values, $N = 1$, 2, 3, 4, 9, 13 and 26, fit the three eclipses of unflagged data without predicting an eclipse where the data show none. Only $N = 13$ and 26, $P = 137.90$ and 68.95 d, also predict the fourth; 3,198 of 3,200 values of N fail one test or the other. The eclipses that only 68.95 d predicts fall in gaps of the TESS coverage, except at the start of sector 101 (Figure 1a). The two-star models fit the eclipses equally well at both periods, so the eclipse shapes cannot choose between them.

Three observations favor 68.95 d. First, flagged cadences at the start of sector 101 rise as the egress of a shallow eclipse, at the time that only 68.95 d predicts (BTJD 4100.898) and at 0.997 times the depth of that kind (Figure 1b). Five stars of similar brightness on the same CCD show no such rise; two brighter ones show ramps at that sector start, and the event was examined after the data were seen. Second, ASAS-3 observed the star every two days from 2000 to 2009 (Pojmanski 1997). On the ten nights that only 68.95 d places in eclipse, a test whose rule was fixed before it was run finds them fainter by 1.04 times the TESS depths, with $p = 0.006$ over 20,000 random shifts of the ephemeris, or 0.012 with no shift excluded (Figure 1c). Each deficit is a single point, so ASAS-3 leans toward 68.95 d without deciding. Third, a NEOWISE visit (Mainzer et al. 2014) on 2014 September 24 covers the deep eclipse that only 68.95 d predicts: HD 170312 fades by 0.13 mag in W2, faintest at the predicted time within the 95 minutes between exposures, while none of 11 stars in the same frames, one as bright, fades by more than 0.02 mag. With no visit at an eclipse both periods predict, the test has no control, and the visit was found after the period analysis, so it favors 68.95 d without establishing it. ASAS-SN photometry of 2014 to 2026 (Shappee et al. 2014; Kochanek et al. 2017; Winecki & Kochanek 2024) is also fainter at the eclipses both periods predict ($p < 5 \times 10^{-5}$), by more than the TESS depths, but does not separate the periods.

The period will be settled by a deeper eclipse at BTJD 4475.8152 ± 0.0002 (2027 March 11), five 68.95 d cycles after the sector 102 eclipse; 137.90 d predicts none then. TESS sector 114 covers that date. The classification rests on photometry: I found no public spectrum of the system, and Gaia DR3 gives it no radial velocity.

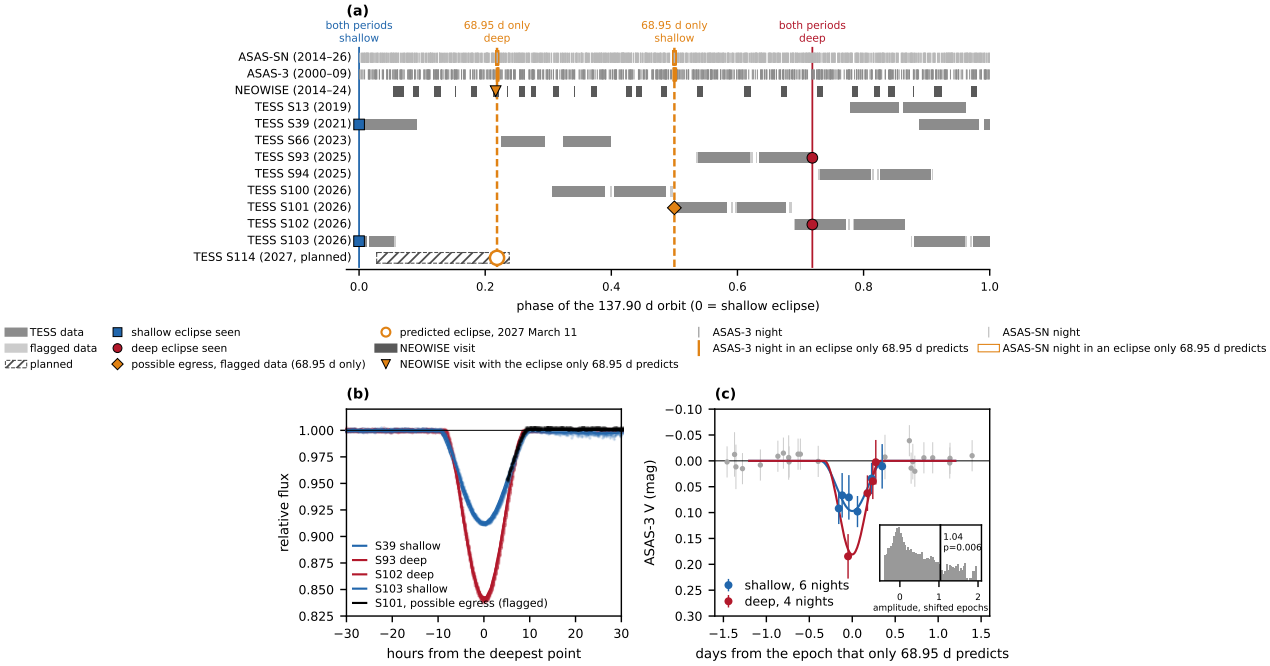


Figure 1. (a) Phases of the 137.90 d orbit (0 at the shallow eclipse) covered by each TESS sector and, in the top rows, by ASAS-SN camera-nights, ASAS-3 nights and NEOWISE visits. Solid lines mark the shallow (blue) and deep (red) eclipses that both periods predict; dashed orange lines, those that only 68.95 d predicts. Orange ticks are the ten ASAS-3 nights of panel (c), the orange triangle the NEOWISE visit at such a deep eclipse, and hollow boxes the ASAS-SN nights in those windows, whose test is undecided. The marks show when data exist, not whether the star was fainter. Markers show the eclipses seen, a possible egress in flagged cadences of sector 101 and the predicted eclipse of 2027 March 11. (b) The four eclipses in SAP flux with 1 h medians, aligned on their deepest point; black, the flagged start of sector 101 at the time that 68.95 d predicts. (c) ASAS-3 nightly V means on those ten nights and, in grey, other nights, with the TESS eclipse shapes; error bars are scaled by 2.9 to the scatter of the data. Inset: amplitudes at 20,000 shifted epochs (log scale), 0.6 percent of which reach the fitted 1.04.

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